
Data Science is Impossible

— Boston University CS 506 - Lance Galletti —

Credibility can be manufactured if you're not aware of the shortcuts your brain is making

How to learn / build knowledge

- I just learned X
- How does X fit in the set of hypotheses that I already have?
- Do any of them conflict with X in what we anticipate?
- Does X conflict with any of them in what we anticipate?
- Do I need to change my prior understanding to make room for X?

Knowledge grows iteratively. Accepting something new means refining or discarding something old.

**If a theory is equally good at explaining
every outcome, it has no knowledge**

**Willingness to refine without
feeling personally threatened**

**Biggest problem: our brains are not
neutral instruments.**

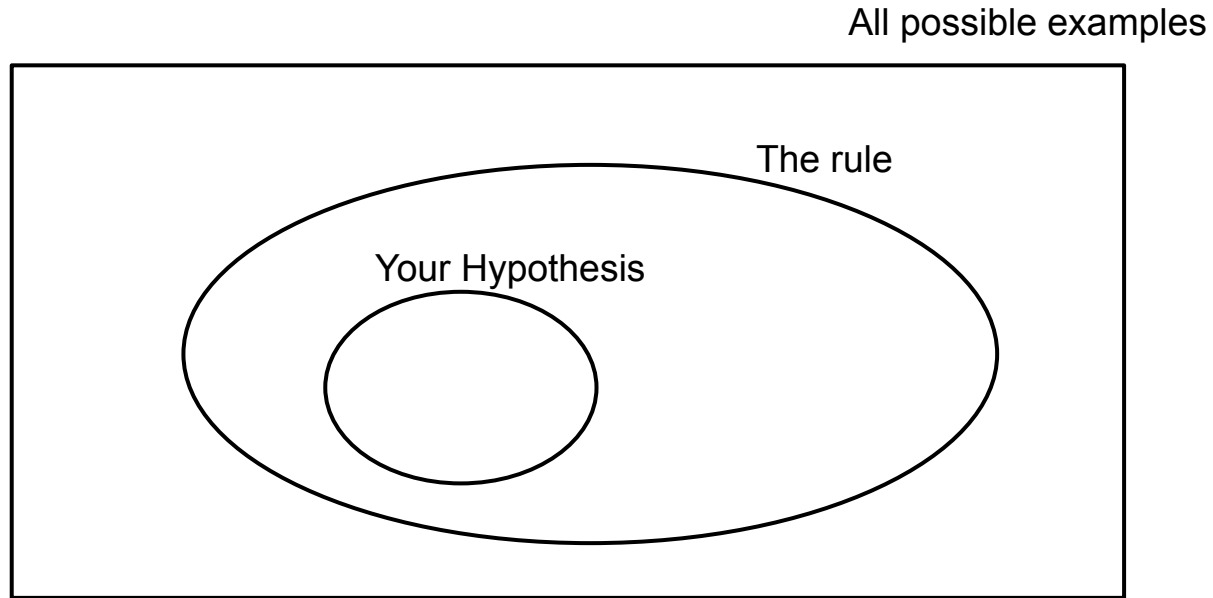
**Humans are imperfect interpreters of
evidence**

Cognitive Biases

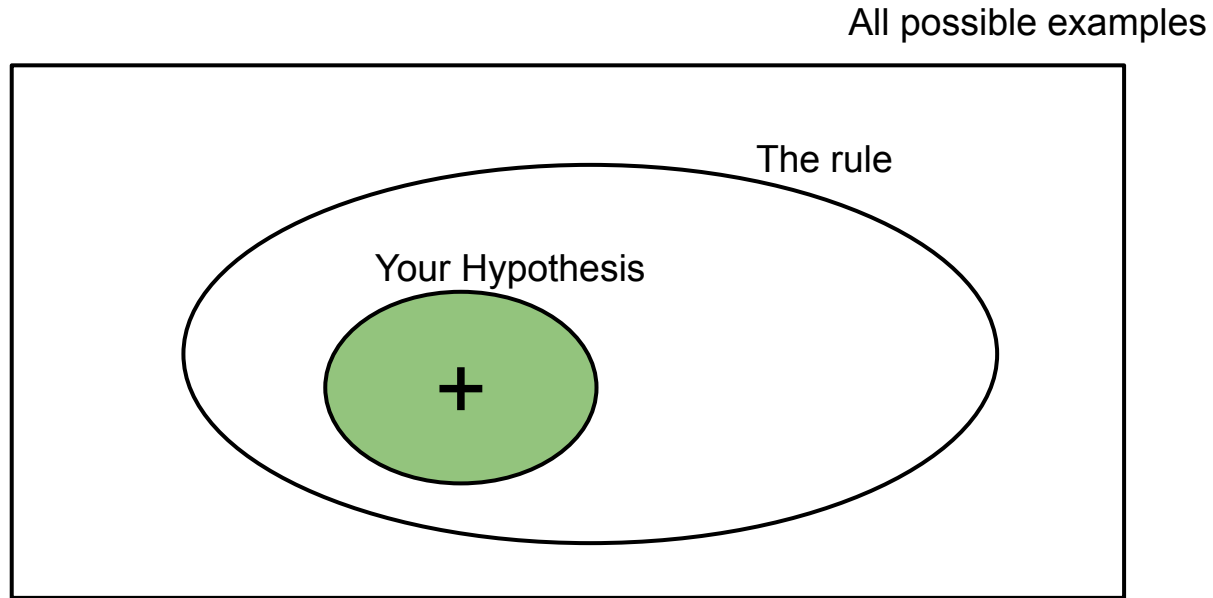
Confirmation Bias

Tendency to favor information that supports what you already believe

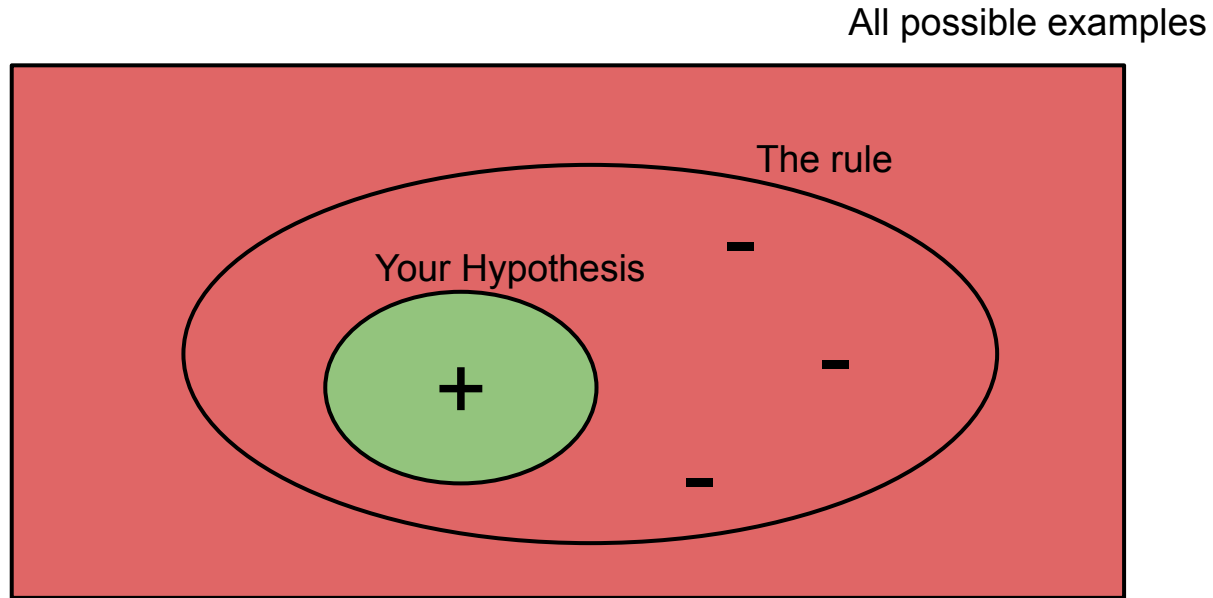
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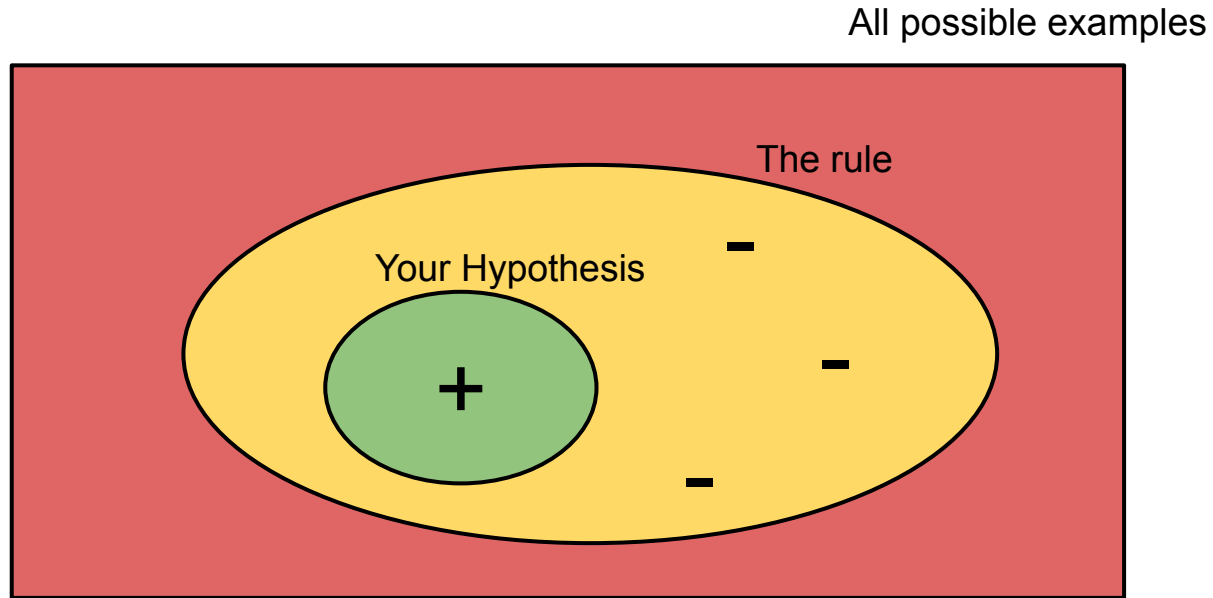
Confirmation Bias



Confirmation Bias

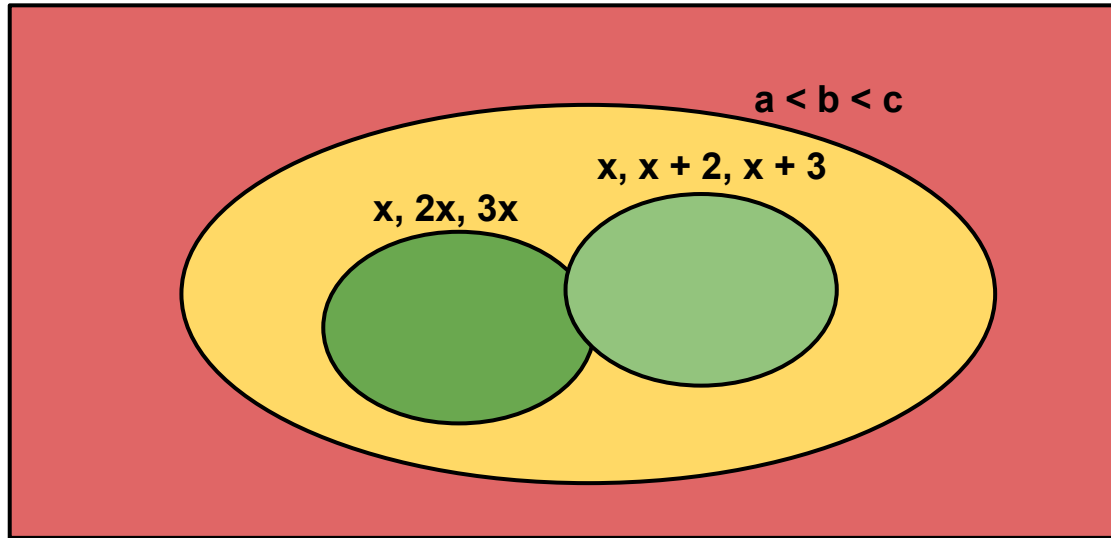


Confirmation Bias



Confirmation Bias

All possible examples



Confirmation Bias

“Am I only noticing evidence that agrees with me?”

“Am I giving this claim a fair chance to be wrong?”

“Could I have interpreted almost any result as supporting what I already believed?”

Availability Bias

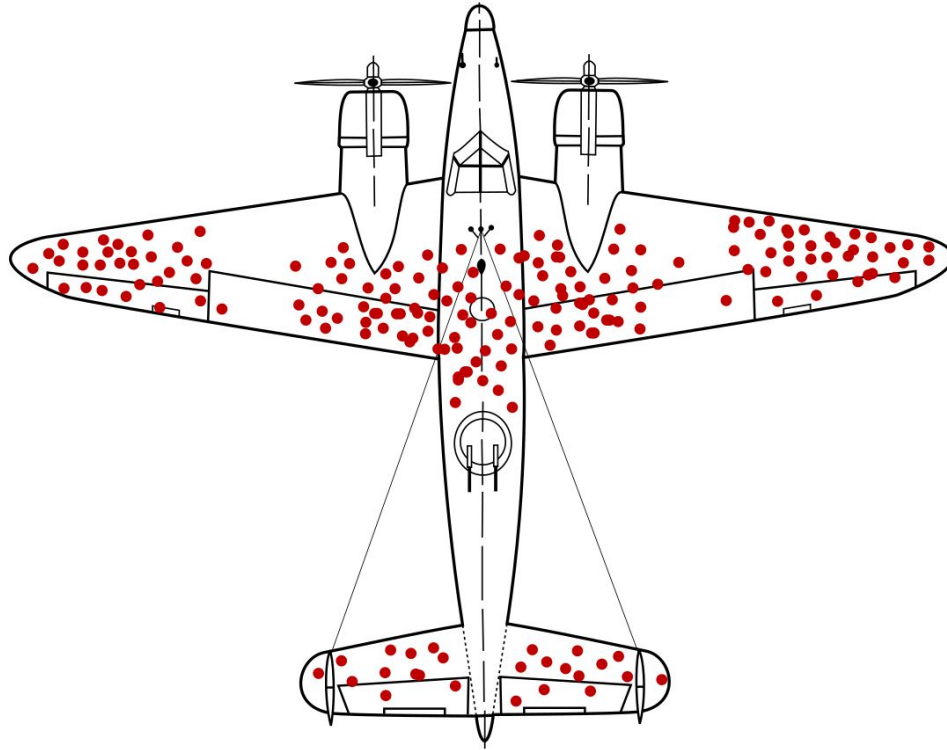
When systems (like social media) repeatedly show you the same kinds of examples, they start to feel representative even when they aren't.

- You see lots of TikToks of airplane incidents in one week and start thinking flying has become much more dangerous, even if the underlying accident rate has not meaningfully changed.

Availability Bias

**“Am I relying on vivid examples
instead of real base rates?”**

Survivorship Bias



Survivorship Bias

“What data might be missing?”

Availability + Survivorship

You see lots of TikToks showing fitness influencers' successful diets and routines, so those approaches start to seem reliably effective. When you try them and they don't work, you conclude something must be wrong with you. But your feed disproportionately shows the people for whom the approach appeared to work, not the much larger number who tried it and got unremarkable or unsuccessful results.

Survivorship: success is more likely to be shown on social media

Availability: success rate is overestimated

Anchoring Bias

Fill out one of the forms at random (don't use AI - that would defeat the purpose)

Anchoring Bias

A laptop is listed as “Originally \$1,999, now \$1,299.” Even if \$1,999 was never a realistic market price, that first number becomes an anchor, so \$1,299 feels like a bargain.

What anchoring bias is not: If I say “last year the exam average was 60%” and you use that number as the expected difficulty of exams this semester.

Starting with prior information is not anchoring bias. Anchoring bias is failing to adjust enough away from a starting point when the evidence says you should.

Anchoring Bias

**“Am I being pulled by the first
number or idea I saw?”**

Hindsight Bias

In spring 1997, snowmelt caused the Red River to rise.

Official forecasts predicted a crest of ~49 feet.

The city built flood defenses (dikes, sandbags) to that level.

The river ultimately crested at 54 feet—far beyond predictions.

“It was obvious the river would exceed forecasts.”

Hindsight Bias

**“Am I treating outcomes as
inevitable after the fact?”**

**“Who of all the Wise could have foreseen it? Or, if they are wise, why should they expect to know it” -
Lord of the Rings**

Framing Bias

90% survival rate sounds different than 10% mortality rate

Framing Bias

“Am I being swayed by wording or presentation?”



The Palantír doesn't create false beliefs; it amplifies the worst possible interpretation of true information at the moment he's least able to resist it.

You don't need false data to mislead someone... just selective data, repeated often enough, shown to a mind that's already leaning one way.

**Beyond cognitive biases:
statistics literacy**

**Where even well-intentioned
humans can be fooled**

No context statistics

A test for a disease is 99% accurate. Is this a good test?

If the disease affects only 1% of people, a test that tells everyone they don't have the disease is 99% accurate... But that's not helpful / useful.

Correlation vs Causation

- Pregnant people are more likely to have cavities
 - Could be hormonal changes or could be different eating habits. Doesn't tell the whole story and you shouldn't jump to conclusions.
- The more ice cream is sold, the more people drown
 - Both increase during the summer. One doesn't cause the other.
- Coffee drinkers have more heart attacks
 - This is from an old study from the 70s where coffee drinkers were more likely to smoke. If you remove the confounding effect of smoking, the effect of coffee on heart attacks vanishes.
- Go to bed with shoes on you'll most likely wake up with a headache
 - Both are related to a third variable: being drunk

Has the world gone coco? Eating chocolate can help you LOSE weight

GOOD news slimmers! New research claims that eating chocolate can actually help you beat the bulge.

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By Laura Mitchell / Published 30th March 2015



CHOCOHOLIC: New research reveals that eating chocolate can actually help you lose weight [GETTY]

It's the diet that everyone has been waiting for.

A German study has found that eating chocolate can reduce your waistline, lower your [cholesterol](#) and help you sleep.

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Dieting? Don't forget the chocolate

By Modern Healthcare | April 11, 2015

The Outliers team digests a lot of food research in the course of a week, but a pair of recent studies that came over the electronic transom stood out for their quirk.

First, according to a study published in JAMA Internal Medicine, an apple a day doesn't necessarily keep the doctor away. That's according to proverb-busting research that found daily apple eaters had just as many doctor visits as those who ate fewer or no apples.

But while apples won't necessarily make you healthier, researchers in Germany seem to have found the answer to a dieter's prayers: chocolate promotes weight loss.

A study out of the Institute of Diet and Health in Mainz found that adults on a strict low-carbohydrate diet who indulged in about 1½ ounces of dark chocolate daily lost 10% more weight than those in another group who were on the same diet minus the treat. The chocolate eaters also maintained their weight loss, while the other group packed the pounds back on.

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Scientists say eating chocolate can help you lose weight

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Monday, March 30, 2015

Adding chocolate to a diet led to the "easiest and most successful weight loss", scientists said.

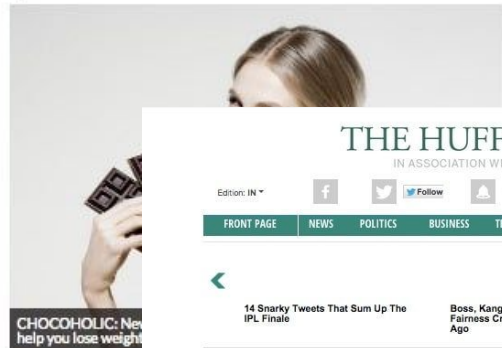


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It's the diet that ever

A German study has lower your cholesterol



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28 May 2015
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Excellent News: Chocolate Can Help You Lose Weight!

ANI

Posted: 31/03/2015 16:21 IST | Updated: 31/03/2015 16:21 IST



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A new research has revealed that chocolate can aid weight loss when combined with a low-carb diet.

Johannes Bohannon, research director of the nonprofit Institute of Diet and Health, said that what is important is the specific combination of foods in your diet when trying to shed those extra pounds, the Daily Express reported.

Bohannon added that just lowering the proportion of carbohydrates is not a reliable

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Name: Government



Pass the Easter Egg! New study reveals that eating chocolate doesn't affect your Body Mass Index ... and can even help you LOSE weight!

- New research from Roy Morgan reveals there's no proof that chocolate consumption affects BMI
- Currently two thirds of Australians eat chocolate at least once a month
- A study from German researchers has also found there's a connection between cocoa diets and increased weight loss
- Chocolate also found to benefit brain, heart and stress levels

By SAM BAILEY FOR DAILY MAIL AUSTRALIA

PUBLISHED: 01:22 EST, 31 March 2015 | UPDATED: 16:14 EST, 31 March 2015

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From the endless chocolate blocks passed around the office, to the glaring supermarket aisles and the family relatives who miraculously appear with baskets of eggs, Easter can be a minefield to navigate if you're trying to watch your waistline.

But according to new research, there's no need to go easy on the eggs this week, with a Roy Morgan study revealing there is no direct connection between chocolate consumption and an increasing Body Mass Index (BMI).

This should come as sweet relief for chocoholics when according to Roy Morgan, two thirds of Australians admit to munching on chocolate at least once a month.

Scroll down for video



Eggcellent news: A chocolate a day is found to not affect your Body Mass Index



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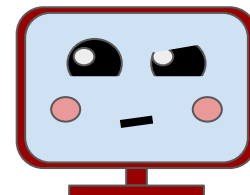
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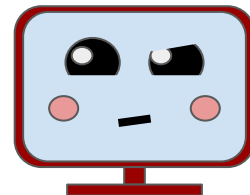


The Study

5 men and 11 women (16 total in the study)

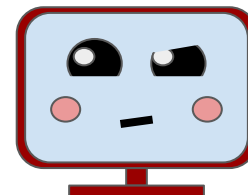
- One group followed a low-carbohydrate diet
- Another followed the same low-carb diet plus a daily 1.5 oz. bar of dark chocolate
- The rest, a control group, were instructed to make no changes to their current diet

They measured 18 different things over 21 days including blood pressure, sodium, sleep quality, wellness, and weight.



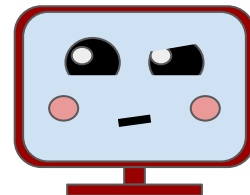
The Result

Those in the chocolate group lost weight 10% faster than the diet group. Both diet groups lost more weight than the control group.



Why it's not real

The experiment is set up to find a pattern.



Simpson's Paradox

	applied	accepted	rate	applied	accepted	rate
TOTAL	1184	274	23%	2470	584	24%

Simpson's Paradox

	applied	accepted	rate	applied	accepted	rate
Computer science	26	7	27%	228	58	25%
Economics	240	63	26%	512	112	22%
Engineering	164	52	32%	972	252	26%
Medicine	416	99	24%	578	140	24%
Veterinary Med	338	53	16%	180	22	12%
TOTAL	1184	274	23%	2470	584	24%

Data Science is Impossible

- You'll never know for sure if you're right. You can only gather evidence.
- The observer (you) affects things too (bias)

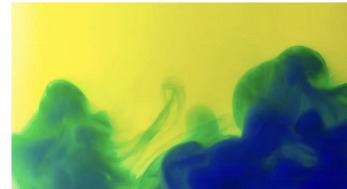
OpenAI claims huge maths breakthrough on a famed 'Millennium Problem'

The company says it has cracked the physics of fluids using artificial intelligence, solving one of the toughest puzzles in mathematics.

By [Davide Castelvecchi](#)

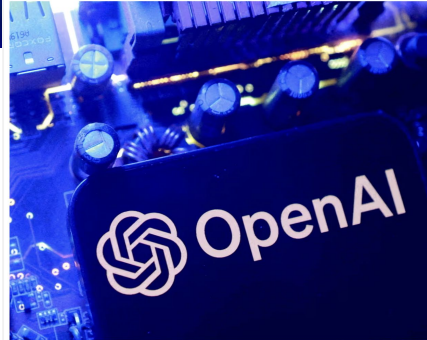


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OpenAI claims to have solved maths problem that stumped humans for decades

Company behind ChatGPT says 10,000 of its AI systems cracked the Navier-Stokes problem in 88 hours



OpenAI denied that the company had used work from rivals to solve the notoriously difficult puzzle. Photograph: Dado Ruvic/Reuters

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The AI Researcher Who Just Quit Anthropic Says It's 'Crunch Time for Humanity'

Jacob Coxon talks to WIRED about the "mini Manhattan project" inside Anthropic, the problem with alignment, and why AI labs have just a few years left to make their systems safe.



Evan Hubinger ✓

@EvanHub

Jacob is correct here—we really do earnestly believe AI could kill all humans! I personally think it is >10% within the next decade. I believe Anthropic is trying its best, but we do not yet have a plan to solve alignment for superintelligence and are not clearly on track to.



Jacob Coxon ✓ @hilbertspaess · 8h

Replying to @hilbertspaess

The people building AI earnestly believe that it could kill us all by the end of the decade. This is not a marketing stunt. If anything, many executives and senior researchers will couch their phrasing in the press to sound sensible - but I hear the same people express fear

9:27 PM · Sep 8, 2026 · 10M Views



https://www.claymath.org/news/navier-stokes-announcement/?utm_source=chatgpt.com

What do all these articles agree on?

Why AI Safety?

- Cybersecurity -> <https://youtu.be/u6nvJkpZzuw> (stuff I've been working on)
- Misinformation
- Job displacement
- Autonomous weapons
- Agency

Agency

Social Media feeds decide what we see and don't see

Ads might shape our wants and needs

Political Campaigns might shape our values

All try to better understand us so they can better influence us. AI is a step beyond that. How do you interact with AI?

Agency

What starts as just daily consultation of AI

- What's the weather like and should I wear today?
- What watch should I buy?
- What kind of savings account should I have?
- Where should I go on vacation?

Agency

Gradually can change into:

- Should I break up with my partner?
- What should be my major?
- How many kids should I have?
- Where should I live?
- Who should I vote for? -> maybe the end of democracy?

Agency

It doesn't even have to be malicious. And we might still have the illusion that we are the ones making the final decision.

Agency

Now imagine a world where AI's research outpaces our ability to learn it.

Let's say AI does AI research and comes up with a new AI model and asks us to deploy it. We can't really take the time to understand this new model because it would take 100+ years for a human to comprehend fully so we ask for a summary in order to make a decision.

You have to trust that the compressed information you get from AI in order to make a decision is devoid of selection that could sway you one direction. The AI now controls the information you see in that compression.

Agency



Example: Traffic

Sometimes global optimization requires keeping individuals in the dark. If everyone is aware of a shortcut traffic gets worse for everyone. If no one is aware and everyone takes a slightly longer route then everyone gets home faster on average.

Some questions worth thinking about

- Is there value in a human doing something imperfectly if an AI could do it better?
 - Diagnosing a disease
 - Art
- Which decisions, should humans refuse to delegate?
- If future generations prefer a highly AI-mediated life, should our current preference for human agency constrain them?